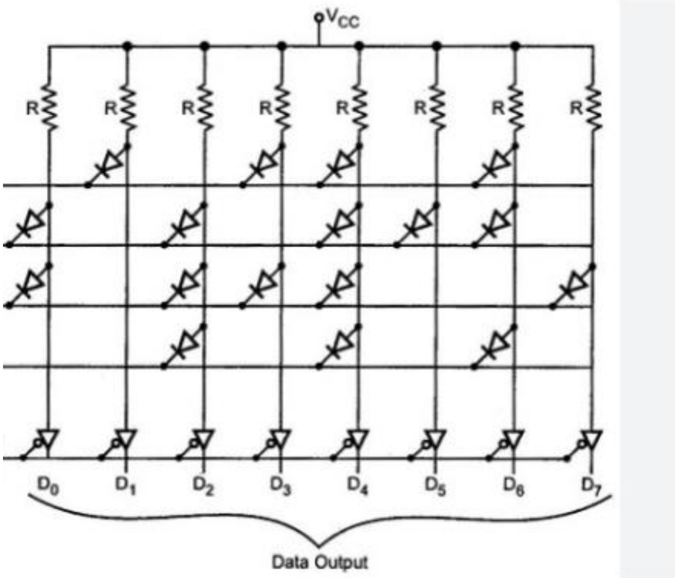
Q1. Explain how to make a read-only memory (ROM) using diodes? **(2 marks)**



Structure

• Inputs A0, A1 are given to a 2:4 decoder

• Decoder activates one word line (row) at a time • Rows → Word lines • Columns → Bit lines (outputs D0 to D7)

• Resistors are connected to VCC (pull-up)

Working (Step-by-Step)

1. Address inputs (A0, A1) are applied to the decoder.

2. The decoder selects one row based on the input:

• 00 → Row 00 active

• 01 → Row 01 active 5

• 10 → Row 10 active

• 11 → Row 11 active

3. The selected row becomes HIGH, others remain LOW.

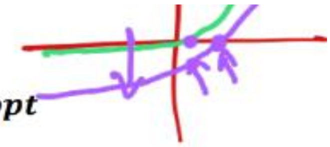
4. Now each column is checked: • If a diode is present at the intersection, it becomes forward-biased and current flows. • Output becomes HIGH (1). • If no diode is present, no current flows, and the output remains LOW (0).

5. All outputs (D0 to D7) together form the stored binary word.

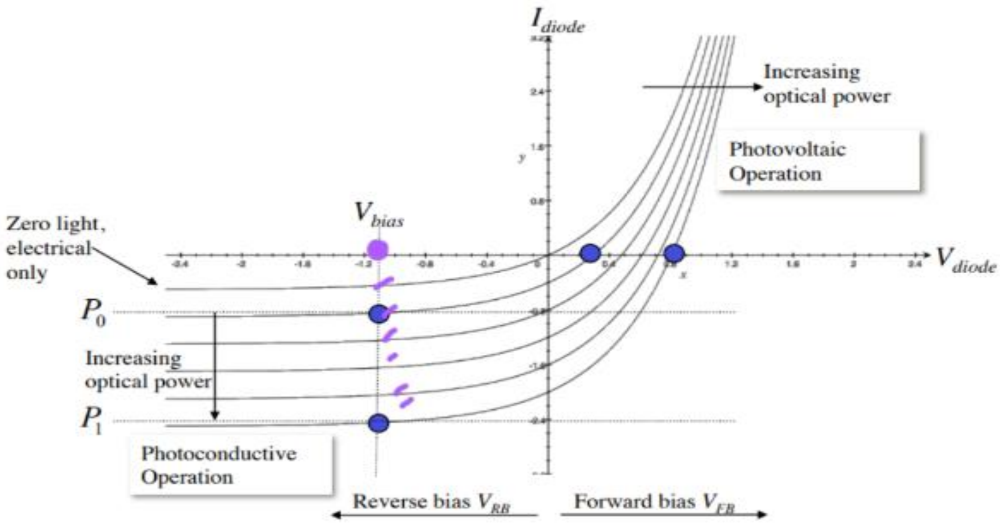
marks only if diode arrangement wrt 1 and 0 is mentioned. If diode is placed for 1 and open circuit for 0 in row/columns: This explanation or figure is enough for full marks.

Q2. Explain how does a photodetector work using band-diagram with direction of electron/hole flow and finally current-voltage plot. **(2 marks)**

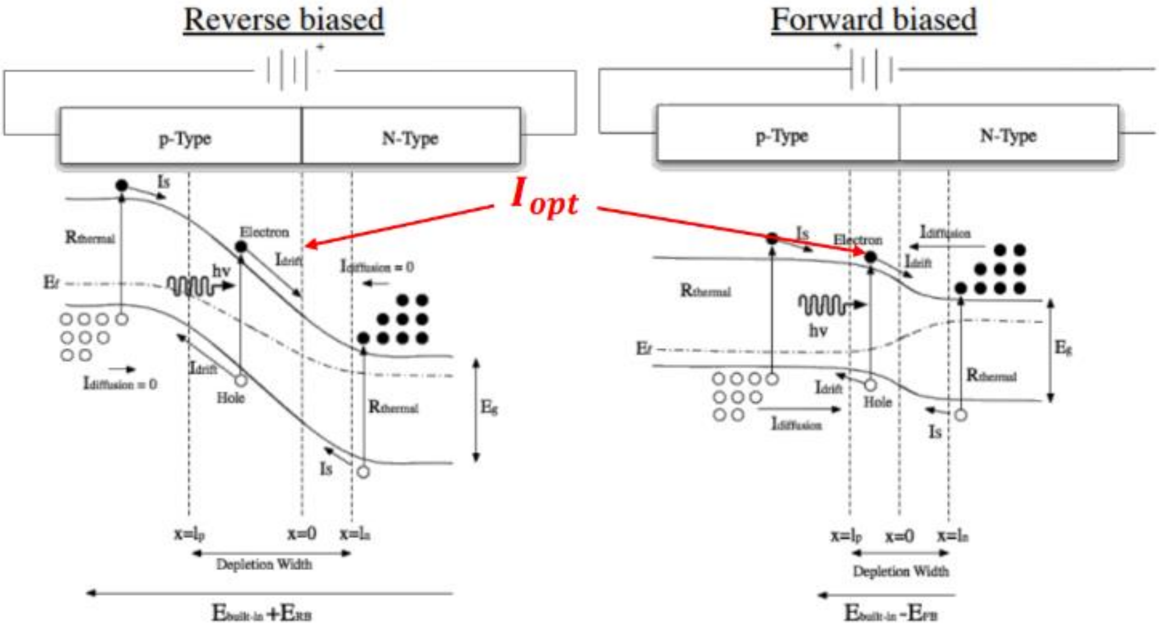
$$I_{Total} = I_0 \left(e^{\frac{V_A}{V_T}} - 1\right) - I_{opt}$$



Impact of e-hole pair generated due to optical source



Impact of e-hole pair generated due to optical source



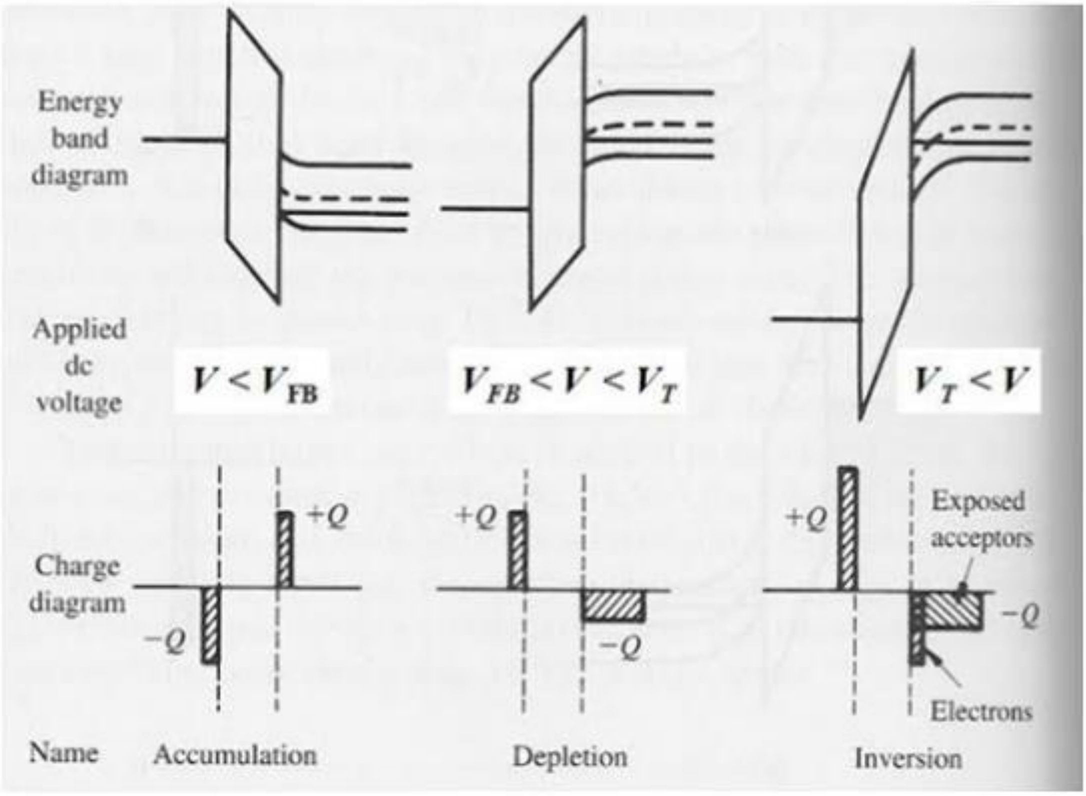
Q3. Explain the operation of MOS Capacitor with P- type semiconductor using the energy band diagram and charges plot for each mode operation. **(6 marks)**

a) Accumulation mode. **(2 marks)**

b) Depletion mode. **(2 marks)**

c) Strong Inversion mode. **(2 marks)**

We have considered VFB= 0 for the following diagrams.



1 mark for BD and 1 mark for charge plot